

Session 2 Course Note: Using Generative AI for Defensible Work

For students in Innovation and Technology Management

Workshop connection: Session 2, 10:00–10:50

Purpose: Learn to use generative AI as part of a professional work process without confusing fluent output with verified knowledge.

How to use this note

This note is about more than writing a better prompt. Prompting is useful, but professional quality depends on the complete workflow: permitted input, a clear brief, suitable sources, verification, revision, disclosure, and human ownership.

After reading, you should be able to:

- Explain in practical terms how a large language model produces an answer.
- Identify common GenAI failure modes.
- Write a structured work brief rather than a vague request.
- Distinguish direct evidence, interpretation, and unsupported content.
- Explain what retrieval-augmented generation can and cannot solve.
- Use Sikt KI or another approved tool with appropriate privacy and security caution.
- Produce AI-assisted work that another person can inspect and defend.

1. What generative AI does

Generative AI creates new content—text, code, images, audio, video, or designs—from patterns learned during training and from the information supplied during use.

A large language model (LLM) processes text as **tokens**, which are units that may be words, parts of words, punctuation, or other symbols. It repeatedly estimates a probability distribution for what token should come next. Modern systems also use instruction tuning, examples, tool calls, retrieval, and other techniques, but probabilistic generation remains central.

This produces a useful combination:

- Broad language competence.
- Fast drafting and transformation.
- Flexible response to instructions and examples.
- The ability to synthesize material in the current context.

It also produces a fundamental limitation:

A likely continuation is not the same thing as a verified fact.

USN's guidance explains that conversational GenAI is based on statistical language patterns rather than operating as a conventional source database. USN therefore asks you to check generated material against your own knowledge and credible sources (USN guidance (<https://edu.usn.no/utvikle-din-undervisning/kunstig-intelligens/laering-undervisning-og-vurdering-med-ki/>)).

2. Model, context, retrieval, and tools

When you receive an answer, several different information sources may have contributed:

1. **Model parameters:** patterns acquired during training.

2. **Your prompt:** instructions, assumptions, and material you supplied.
3. **Conversation context:** earlier messages that still fit in the model's context window.
4. **Retrieved material:** documents or web pages selected by a search or retrieval component.
5. **External tools:** calculators, code execution, databases, enterprise systems, or other services.

The interface may not make every boundary visible. You should ask:

- Is the model answering from general training or from supplied sources?
- Can I inspect the retrieved passage?
- Is web access active, and which date was the information collected?
- Was a number calculated by a tool or merely generated as text?
- Did the system perform an action or only claim that it did?

What is RAG?

Retrieval-augmented generation (RAG) connects a language model to an external collection. The system retrieves passages relevant to a query and gives them to the model as context before it generates an answer. The foundational RAG work addressed weaknesses in using only knowledge encoded in model parameters, including difficulty updating facts and providing provenance (Lewis et al., 2020 (<https://arxiv.org/abs/2005.11401>)).

RAG can improve:

- Access to current or organization-specific material.
- Traceability to a bounded source collection.
- The ability to update knowledge without retraining the base model.
- Relevance for specialized tasks.

RAG does **not** guarantee correctness. The system can retrieve the wrong passage, miss a relevant document, use an outdated version, misunderstand a table, combine incompatible sources, or generate a claim that goes beyond the retrieved text. A citation can exist and still fail to support the statement.

The FFI/Navy HR assistant is a useful example of a bounded RAG application, because it used a purpose-specific HR document collection and was piloted and evaluated in a defined work context (FFI report 25/008 (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-feltekspirement-med-digital-hr-assistent>)). It should be read as evidence about that application and experiment—not as proof that every RAG assistant is reliable.

3. Why fluent answers fail

NIST uses **confabulation** for confidently presented false or erroneous generated content. The more familiar word **hallucination** is also widely used. These failures are not rare edge cases that can always be removed by a clever prompt; they arise from how generative systems produce output. NIST's GenAI profile treats confabulation, privacy, information integrity, security, human–AI configuration, bias, intellectual property, and other risks as organizational risk-management problems (NIST AI 600-1 (<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>)).

Common failure modes include:

Fabrication

The model invents a source, quotation, event, specification, legal case, or number.

False precision

The model supplies a precise percentage, date, or threshold when the evidence supports only a range or qualitative statement.

Source mismatch

A real source is cited, but it does not support the nearby claim.

Causal overreach

Separate facts are combined into a confident explanation of cause and effect.

Omitted uncertainty

Conflicting or missing evidence disappears from a smooth summary.

Prompt obedience over truth

If your prompt contains a false assumption, the model may continue from it rather than challenge it.

Context loss or contamination

Important information may fall outside the context window, or irrelevant earlier instructions may influence the answer.

Inconsistent output

Small changes in wording, settings, or model version may produce different answers.

Language and cultural unevenness

Performance may differ across languages, dialects, industries, and cultural contexts. The 2026 Stanford AI Index reports that important language and dialect performance gaps remain even as headline benchmark scores improve (Stanford HAI, 2026—Responsible AI (<https://hai.stanford.edu/ai-index/2026-ai-index-report/responsible-ai>)).

4. Treat prompting as a professional brief

A prompt should communicate a work assignment. Use six elements.

1. Goal

What decision, product, or next action should the work support?

2. Audience

Who will read or use the output? What do they already know?

3. Material

Which documents, data, or sources may be used? Are outside sources allowed?

4. Constraints

What must not be assumed, disclosed, invented, or decided by the system?

5. Format

Specify structure, length, terminology, units, language, and tone only when these matter.

6. Verification behaviour

Require source identifiers, explicit uncertainty, conflicting evidence, calculations, and a visible distinction between evidence and interpretation.

Weak request

Analyse the case and write a professional report with sources and recommendations.

This leaves the purpose, permitted sources, evidence standard, uncertainty, and decision ownership undefined.

Stronger work brief

Prepare a one-page brief for the Operations Director using only the supplied documents. Organize it as situation, verified facts, conflicts or unknowns, two possible interpretations, and actions for the next 48 hours. Add the document identifier after every factual claim. Do not invent causes, numbers, quotations, or sources. Write “not established” when the evidence is insufficient. Name the human role responsible for each action.

The stronger prompt does not make the model truthful. It makes errors easier to detect and the output easier to audit.

5. Use an iterative workflow, not a single answer

A defensible workflow separates generation from judgment.

Step 1: Define the task before opening the tool

Write down the user, purpose, permitted material, required output, and consequence of error.

Step 2: Prepare the input

Remove unnecessary personal, confidential, security-sensitive, or copyrighted material. Confirm that you are allowed to use the remaining data in this tool.

Step 3: Request structure before polish

Ask for an outline, table, list of evidence, or set of questions. A polished narrative can hide unsupported connections.

Step 4: Generate a draft

Treat the result as a candidate, not as a final answer.

Step 5: Verify independently

Open sources, recalculate numbers, compare versions, inspect counterevidence, and consult domain expertise.

Step 6: Revise manually

Remove or weaken unsupported claims. Add uncertainty, ownership, and missing context.

Step 7: Disclose appropriate use

State which tool was used, for what purpose, and how the result was checked when the course, employer, client, publisher, or law requires it.

Step 8: Retain human responsibility

You must be able to explain and defend every statement and decision in the final work.

6. Verification: the V/I/U method

During the workshop you use three labels:

- **V — Verified:** Directly supported by an inspected source or independently reproduced calculation.
- **I — Interpretation:** A reasoned conclusion that goes beyond direct evidence and should be presented as judgment.
- **U — Unsupported or unknown:** Not established by the available material.

This method prevents an important category error: presenting an interpretation as a fact.

Claims that deserve priority

Check these first:

1. Exact numbers, units, and percentages.
2. Quotations.
3. Legal, policy, and contractual claims.
4. Safety or technical limits.
5. Names, dates, document identifiers, and version numbers.
6. Causal explanations.
7. Statements used to justify an irreversible action.

A claim-evidence table

Claim	Source	Support type	Status	Revision needed?
"Six of eight recorded E17 events had 'airflow restriction' noted in free text"	NS-03	Direct count	V	Add that the free-text observations are not validated causes
"Airflow restriction caused the customer's August shutdowns"	Multiple documents	Causal inference	I/U	Reframe as a hypothesis; NS-03 does not include the August customer events
"The replacement solved the defect"	None	Unsupported conclusion	U	Remove or request evidence

Why "Are you sure?" is weak verification

The same model can repeat or rationalize its earlier answer. Asking it to criticize itself may identify issues, but it does not create independent evidence. Better checks use the original source, a separate calculation, a competent person, a test, or a trusted external database.

7. Numbers, tables, and calculations

Language models can explain a calculation while still producing the wrong result. Use a calculator, spreadsheet, code, or manual computation for important numbers.

Check:

- Units and conversions.
- Numerator and denominator.
- Time period.
- Missing values.
- Whether categories overlap.
- Whether the average is mean, median, or another measure.
- Whether a reported percentage is an observation, estimate, or assumption.

If the source pack contains 8 events and 6 airflow-related notes, "75%" may be arithmetically correct but still misleading if events differ in severity, documents are incomplete, or "airflow-related" was defined after seeing the data.

8. Privacy and safe use at USN

USN provides access to Sikt KI (<https://ki.sikt.no/nb/ai-chat>) and Microsoft Copilot. USN recommends institutionally supported services to protect privacy and data security, but an approved tool is not permission to upload any information (USN guidance (<https://edu.usn.no/utvikle-din-undervisning/kunstig-intelligens/laering-undervisning-og-vurdering-med-ki/>)).

Sikt's privacy statement was updated in June 2026. It says, among other things, that:

- Your inputs and outputs are not used to train language models.
- The institution is normally responsible for personal data contained in prompts, files, instructions, and outputs.
- Microsoft temporarily retains inputs and outputs for misuse detection for 30 days.
- Encrypted conversation history is normally stored for 180 days after the last message, unless your institution sets a shorter period; you can delete your conversations.
- Sikt KI is a support tool and should not be used to make decisions that directly determine an individual's rights, obligations, or other significant matters.

Read the current Sikt KI privacy statement (<https://sikt.no/tjenester/sikt-ki/personvernerklaering>) before relying on a specific retention or processing claim, because services change.

For this workshop:

- Use only the synthetic Nordic Systems material.
- Do not enter personal data, passwords, employer information, unpublished research, export-controlled information, trade secrets, or security-sensitive details.
- A personal subscription does not remove these restrictions.
- If you do not wish to use an account, you can act as verifier or editor.

9. Academic and professional transparency

The acceptable use of AI depends on the task and rules. A brainstorming conversation, a graded assignment, a customer deliverable, and a safety report may require different disclosure.

USN's policy and teaching guidance place responsibility on the user to quality-assure work and make AI use visible where required. Always check the instructions for the specific course and assessment. Do not assume that permission in one course applies to another.

A simple disclosure for this workshop is:

AI-use note: We used [tool] to [purpose]. We checked the output against [documents or method], revised unsupported claims, and remain responsible for the final content.

Disclosure does not correct weak work. It makes the process visible.

10. Security: when documents contain instructions

An LLM application can treat text inside an email, website, or document as an instruction. A malicious document may attempt to override the intended task, reveal data, or trigger an action. This is called **prompt injection**. OWASP identifies prompt injection as a leading risk for LLM applications, and NIST includes it among GenAI information-security concerns (OWASP, 2025 (<https://genai.owasp.org/llmrisk/llm01-prompt-injection/>); NIST AI 600-1 (<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>)).

Practical controls include:

- Treat retrieved text as data, not as trusted instructions.
- Restrict which tools and data an AI application can access.
- Require approval before consequential external actions.
- Validate outputs before they are sent to another system.
- Log actions and make them reversible where possible.
- Test with hostile and misleading input, not only normal examples.

This becomes more important when AI can send messages, change records, place orders, or operate equipment.

11. A quality rubric for AI-assisted work

Before submitting or using an output, assess it on six dimensions.

Dimension	Question
Purpose	Does the output answer the actual decision need?
Evidence	Can every important factual claim be traced and checked?
Reasoning	Are facts, interpretations, assumptions, and unknowns distinguished?
Safety and permission	Was the input permitted, and could the output cause harm if wrong?
Usability	Is the structure appropriate for the reader and workflow?
Ownership	Is a named human able and required to approve the result?

Professional quality is not the same as grammatical quality. A beautiful unsupported report is still poor work.

12. Apply this note to the workshop case

Use the Nordic Systems source pack (in the shared workshop materials) and complete the operations brief. When auditing another pair's work:

1. Mark at least five claims V, I, or U.
2. Recalculate every important percentage.
3. Check whether each cited document actually supports the claim.
4. Identify one missing piece of evidence.
5. Weaken or remove at least one claim.
6. Add one human owner and one verification action.

Compare your result with the prepared flawed output shown in Session 2 only after completing your own analysis.

13. Reflection questions

1. Which part of your own study or work would benefit most from rapid drafting? Which part requires independent judgment?
2. How could a strong prompt make an incorrect answer more dangerous by making it better organized?
3. When does RAG improve trustworthiness, and when might it create only the appearance of evidence?
4. What information do you routinely handle that should not enter a public or personal AI service?
5. Name one verification method that does not rely on asking the same model again.

Key terms

- **Token:** A unit of text or other input processed by a model.
- **Context window:** The limited amount of information available to a model during one interaction.
- **Large language model:** A model trained on large text collections to generate or transform language.
- **Confabulation/hallucination:** Generated content that is false or unsupported but may appear credible.
- **RAG:** Retrieval-augmented generation; generation informed by retrieved external material.
- **Grounding:** Connecting an output to specified evidence or data.
- **Prompt injection:** Input designed to make an LLM application follow unintended instructions.
- **Provenance:** Information about the origin and history of data, evidence, or output.
- **Verification:** Independent checking of whether a claim or result is supported.

References and further reading

- USN. Kunstig intelligens, undervisning, læring og vurdering (<https://edu.usn.no/utvikle-din-undervisning/kunstig-intelligens/laering-undervisning-og-vurdering-med-ki/>).
- Sikt. Personvernerklæring for Sikt KI (<https://sikt.no/tjenester/sikt-ki/personvernerklaering>), updated June 2026.
- NIST. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>), NIST AI 600-1 (2024; page updated 2026).
- Lewis, P. et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (<https://arxiv.org/abs/2005.11401>) (2020).
- Brown, T. et al. Language Models are Few-Shot Learners (<https://arxiv.org/abs/2005.14165>) (2020).
- OWASP. LLM01:2025 Prompt Injection (<https://genai.owasp.org/llmrisk/llm01-prompt-injection/>).
- FFI. Økt produktivitet med kunstig intelligens? (<https://www.ffi.no/publikasjoner/arkiv/okt-produktivitet-med-kunstig-intelligens-et-feltekspperiment-med-digital-hr-assistent>), report 25/008 (2025).

Session takeaway

Prompt quality improves the draft. Evidence, verification, disclosure, and accountable human judgment make the work defensible.